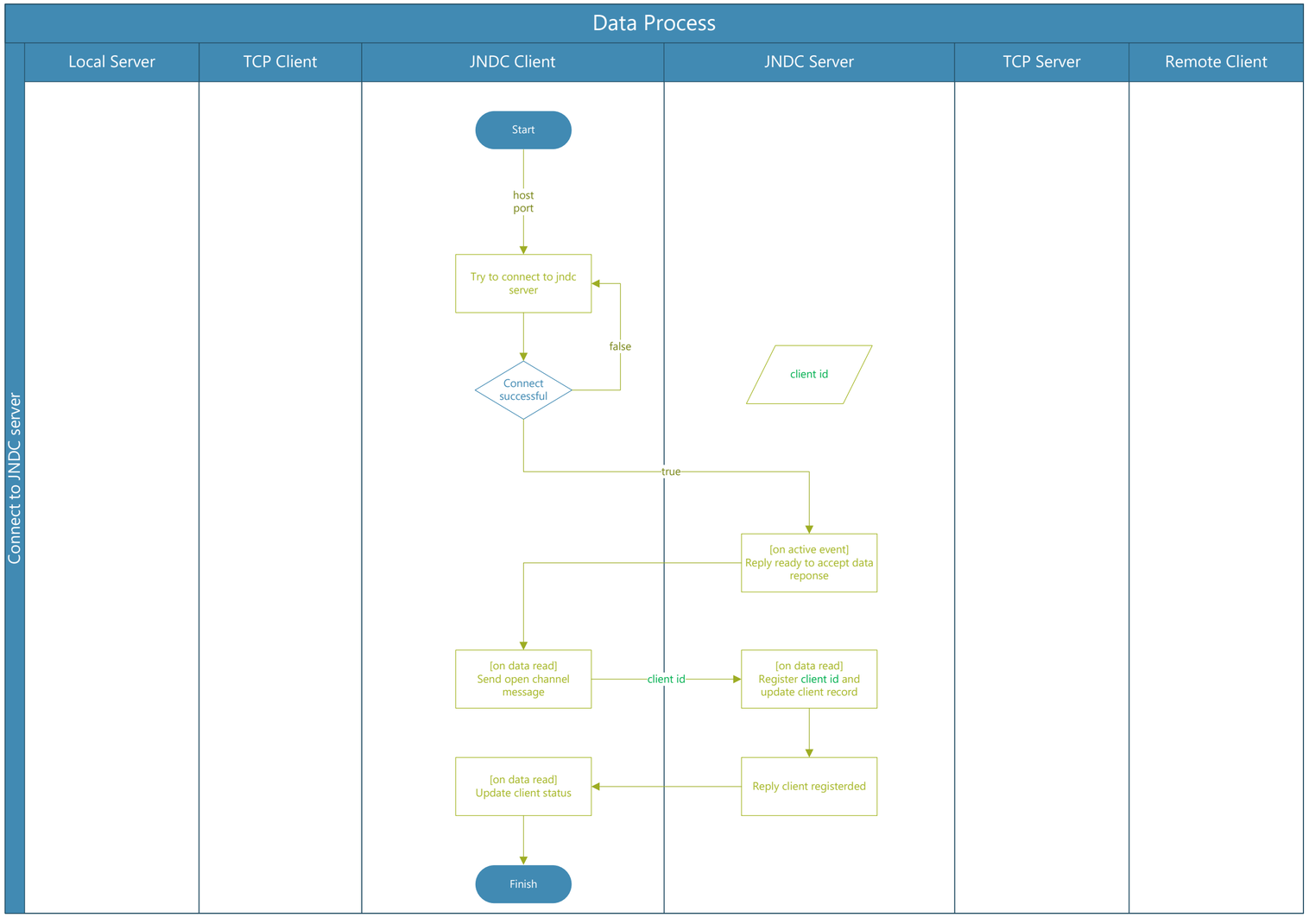
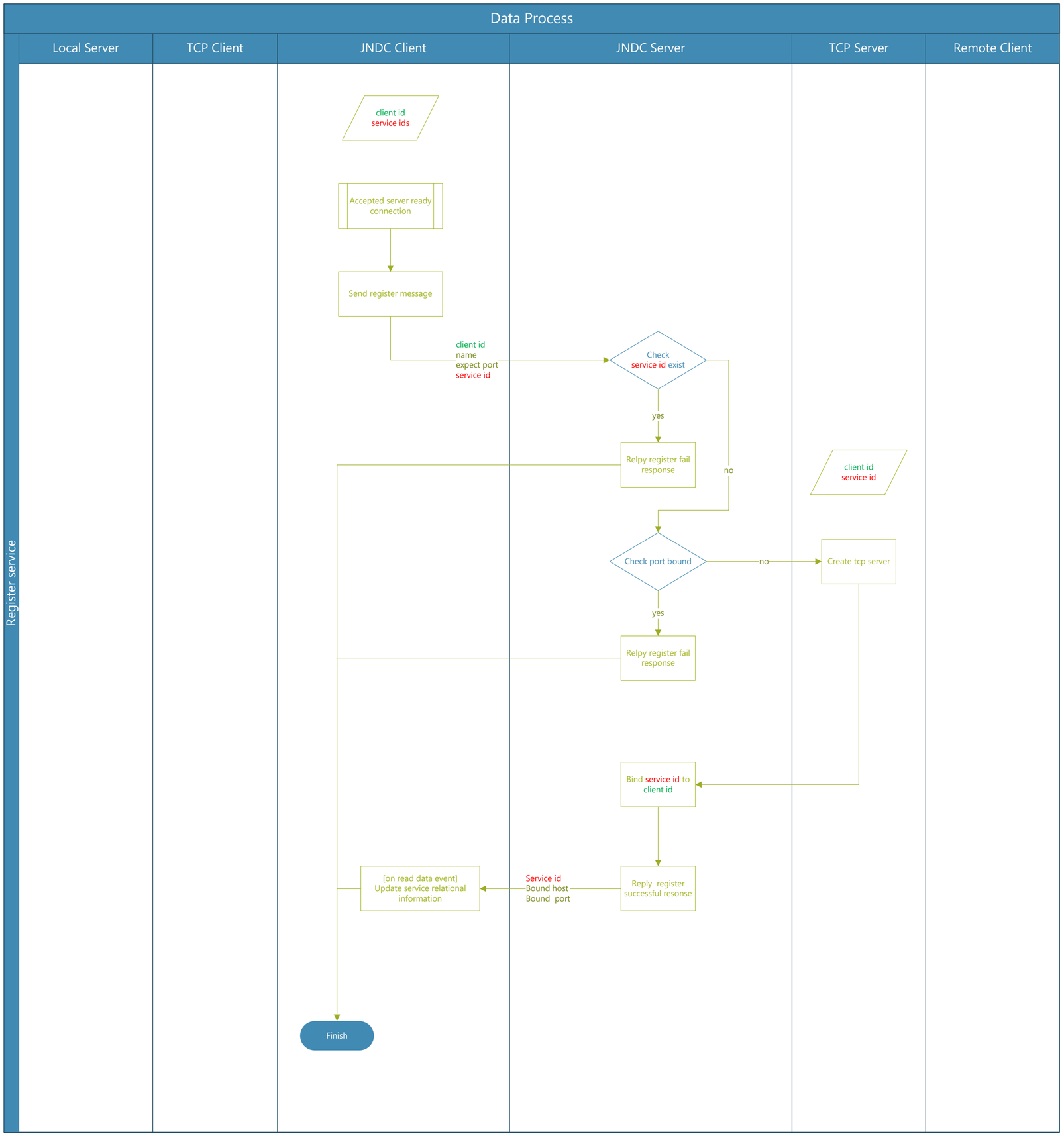
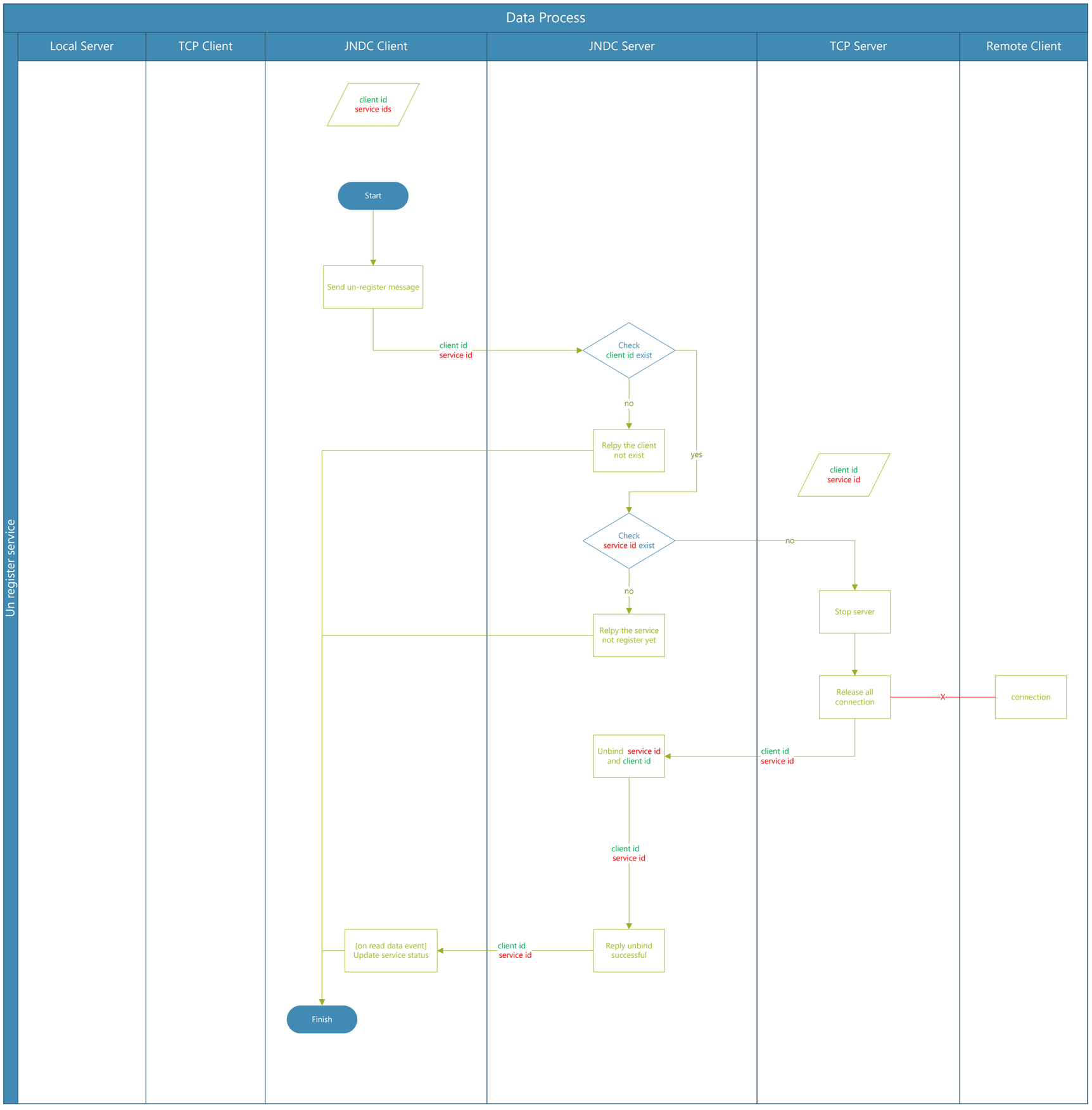
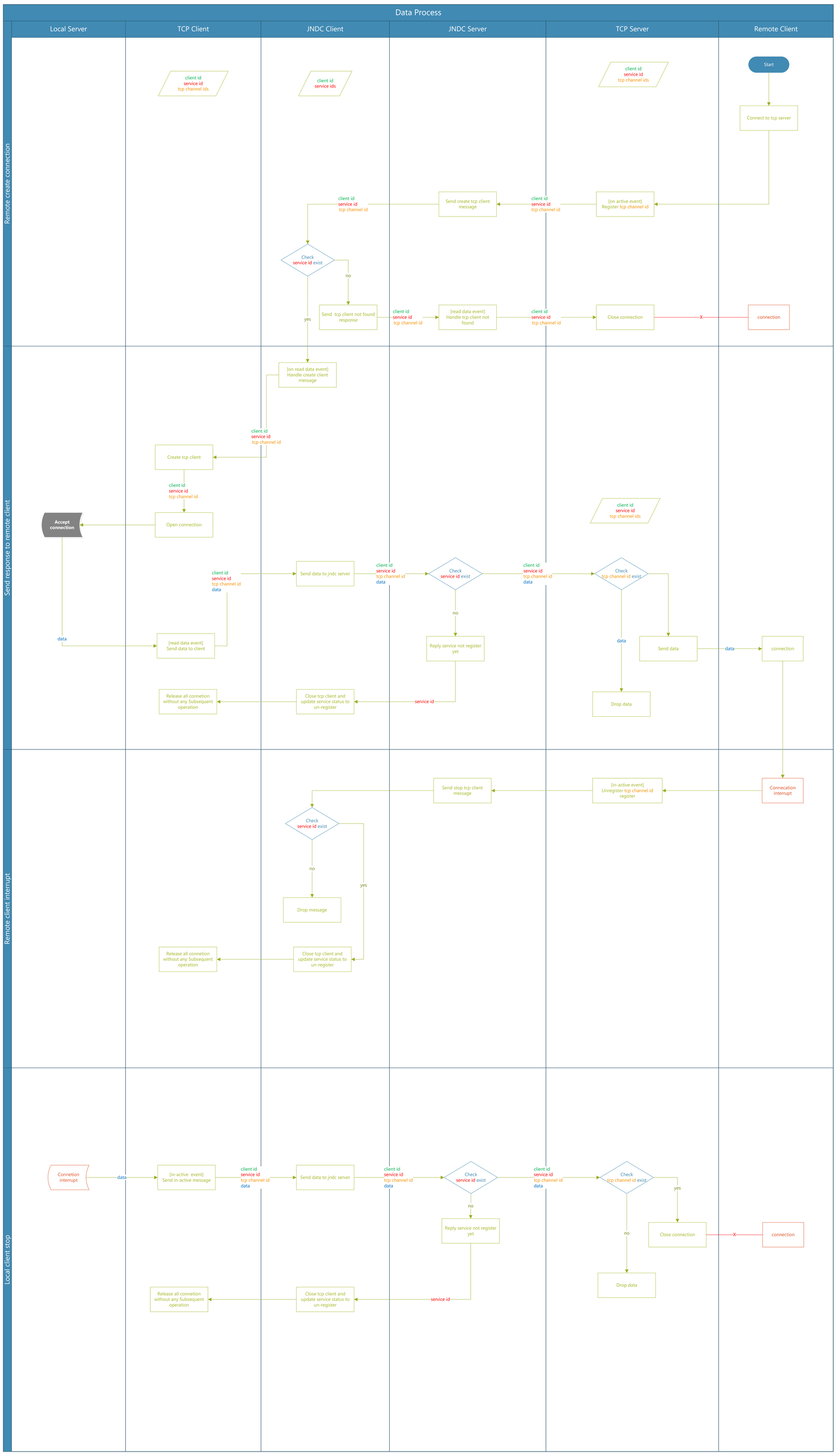
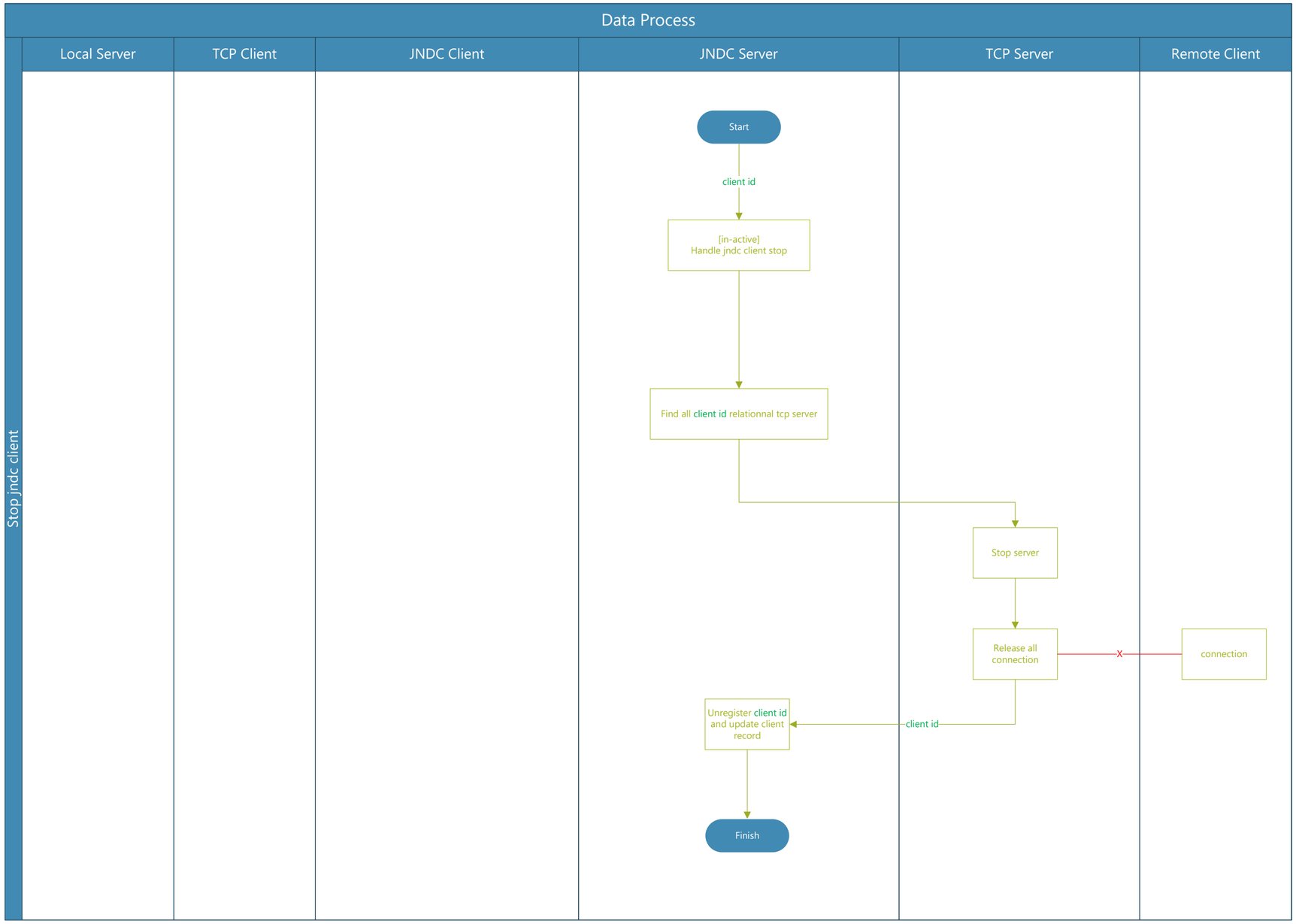
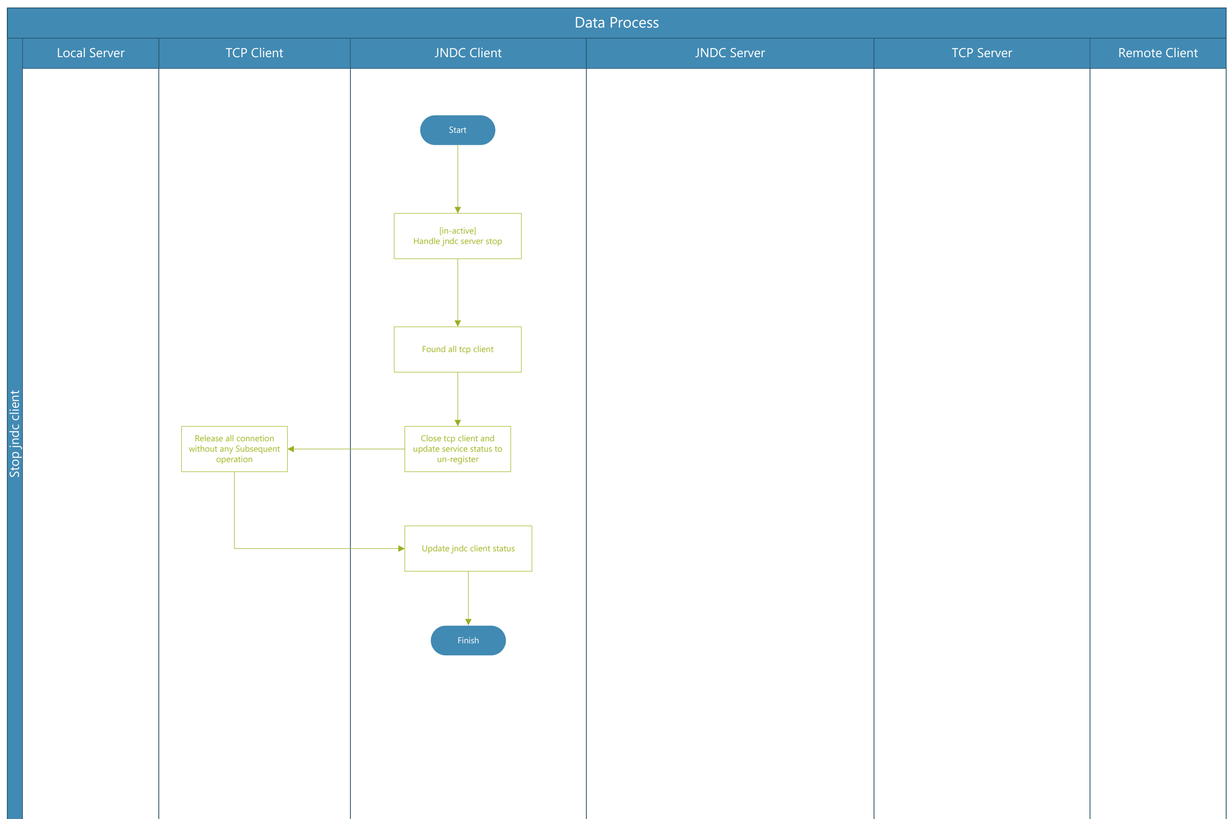

Data Process
Connect to JNDC server
Local Server
TCP Client
JNDC Client
JNDC Server
TCP Server
Remote Client
Start
host
port
Try to connect to jndc server
false
Connect successful
client id
true
[on active event]
Reply ready to accept data reponse
[on data read]
Send open channel message
client id
[on data read]
Register client id and update client record
[on data read]
Update client status
Reply client registerded
Finish


Data Process
Local Server
TCP Client
JNDC Client
JNDC Server
TCP Server
Remote Client
Register service
client id
service ids
Accepted server ready connection
Send register message
client id
name
expect port
service id
Check
service id exist
yes
no
Relpy register fail response
client id
service id
Check port bound
no
Create tcp server
yes
Relpy register fail response
Bind service id to client id
Reply register successful resonse
Service id
Bound host
Bound port
[on read data event]
Update service relational information
Finish


Data Process
Un register service
Local Server
TCP Client
JNDC Client
JNDC Server
TCP Server
Remote Client
client id
service ids
Start
Send un-register message
client id
service id
Check
client id exist
no
Relpy the client
not exist
yes
client id
service id
Check
service id exist
no
no
Stop server
Relpy the service
not register yet
Release all
connection
X
connection
Unbind service id
and client id
client id
service id
client id
service id
[on read data event]
Update service status
client id
service id
Reply unbind
successful
Finish


Data Process
Local Server
TCP Client
JNDC Client
JNDC Server
TCP Server
Remote Client
Remote create connection
client id
service id
tcp channel ids
client id
service ids
Start
Connect to tcp server
client id
service id
tcp channel id
Send create tcp client message
client id
service id
tcp channel id
[on active event]
Register tcp channel id
Check
service id exist
no
yes
Send tcp client not found response
client id
service id
tcp channel id
[read data event]
Handle tcp client not found
client id
service id
tcp channel id
Close connection
X
connection
Send response to remote client
[on read data event]
Handle create client message
client id
service id
tcp channel id
Create tcp client
client id
service id
tcp channel id
Accept connection
Open connection
client id
service id
tcp channel ids
client id
service id
tcp channel id
data
Send data to jndc server
client id
service id
tcp channel id
data
Check
service id exist
client id
service id
tcp channel id
data
Check
tcp channel id exist
no
Reply service not register yet
data
[read data event]
Send data to client
data
Send data
data
connection
Release all connetion without any Subsequent operation
Close tcp client and update service status to un-register
service id
Drop data
Remote client interrupt
Connecation interrupt
[in-active event]
Unregister tcp channel id register
Send stop tcp client message
Check
service id exist
no
yes
Drop message
Release all connetion without any Subsequent operation
Close tcp client and update service status to un-register
Local client stop
Connetion interrupt
data
[in-active event]
Send in-active message
client id
service id
tcp channel id
data
Send data to jndc server
client id
service id
tcp channel id
data
Check
service id exist
client id
service id
tcp channel id
data
Check
tcp channel id exist
yes
no
Reply service not register yet
Close connection
X
connection
Drop data
Release all connetion without any Subsequent operation
Close tcp client and update service status to un-register
service id


Data Process
Stop jndc client
Local Server
TCP Client
JNDC Client
JNDC Server
TCP Server
Remote Client
Start
client id
[in-active]
Handle jndc client stop
Find all client id relationnal tcp server
Stop server
Release all connection
connection
client id
Unregister client id and update client record
Finish


Data Process
Stop jndc client
Local Server
TCP Client
JNDC Client
JNDC Server
TCP Server
Remote Client
Start
[in-active]
Handle jndc server stop
Found all tcp client
Close tcp client and update service status to un-register
Release all connetion without any Subsequent operation
Update jndc client status
Finish